

MEMO TO:

Mr Peet Schutte
M.Sc High Voltage Electrical Engineering
Team Lead – Earthing, Electromagnetic
Fields and Insulation Coordination
Transmission & Distribution Technology
DNV GL - Energy Advisory

MEMO NO: OTD150-0 Rev 2.0**FROM:** DNV GL Energy Advisory**DATE:** 8 Jul 2020**PREP BY:** Dr Pieter H Pretorius**COPIED TO:**

**TITLE: PLAN OF APPROACH: ANTICIPATED 0,4 μ T MAGNETIC FIELD CONTOUR
ASSOCIATED WITH THE NEW OOSTEIND 150 kV SUBSTATION**

1. INTRODUCTION**1.1 Background**

TenneT is currently addressing the positioning and location of the new 150 kV Oosteind substation. The existing 150 kV high voltage station does not meet the 100 MW criterion and for this reason, a new high voltage station needs to be built adjacent to the existing substation.

The existing 150 kV switch yard will be decommissioned but the three existing Enexis 150 kV / 10 kV transformers will remain and will be reconnected at the new 150kV switch yard.

For the exact position of the new station, a magnetic field zone calculation is required in order to ensure that sensitive zones are not affected when falling within the 0,4uT zone. The location of the new substation may “slide” in a Northwest to Southeast line. The exact position is not yet certain and will be firmed by the findings from the magnetic field study.

DNV GL – Energy Advisory was approached for technical support with the magnetic field study. The detailed plan of action (Plan van Aanpak) for the magnetic field study is presented in this document.

1.2 Project Status

The project is in its initial stages with the Plan of Action covered in this document and the development of the numerical model and study in progress.

2. SCOPE OF WORK**2.1 Relevant Project**

The relevant project is referred to as the: Magnetic field study of the new 150 kV Oosteind Substation.

2.2 Project Components

The project comprises of the following components (see Figure 1) with highlighted areas that will be included in the study:

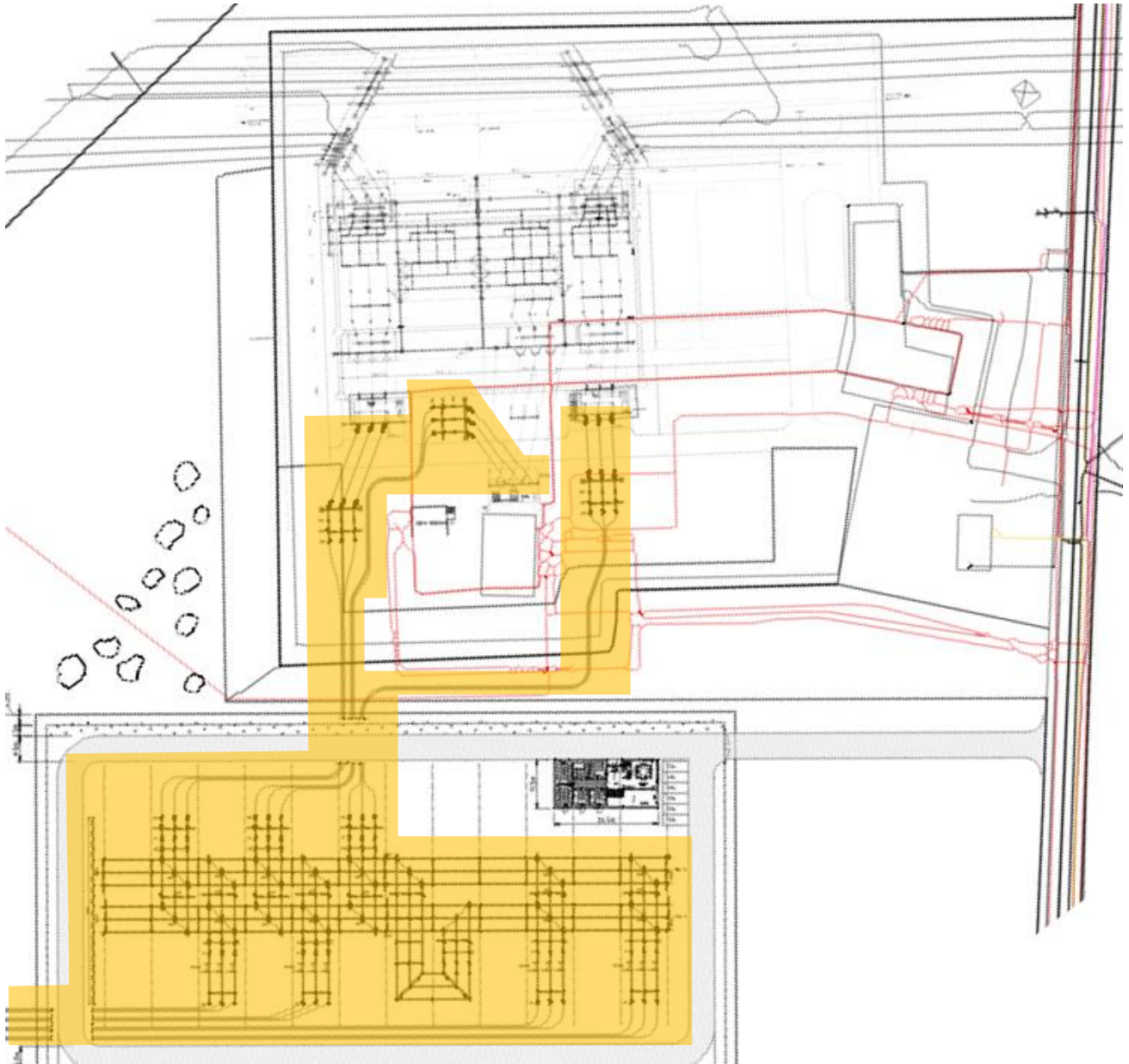


Figure 1: Layout of the new and existing 150 kV Oosteind substation; The components highlighted will be included in the magnetic field study.

2.3 Objective

The objective with this study is to indicate the location of the 0,4 μT zone ("the specific magnetic field zone" [1]) in the vicinity of the new 150 kV Oosteind substation based on the noted energisation and basic assumptions.

It is specifically noted that: The specific magnetic field zone calculated in this report is not the actual magnitude of the magnetic field at a given location at any given time, but rather, it is representative of a magnetic field zone that is based on the calculation approach and assumptions presented in this report, as guided by the policy and agreements related to magnetic field calculations of high voltage cables and substations of the national government [1].

2.4 Tasks to be Executed

In view of the objective, the following technical deliverables are offered:

- Reporting on the substation energization (with basic assumptions) and top view of the associated 0,4uT zone.

3. PROPOSED METHODOLOGY

3.1 Proposed Approach

The magnetic field zone study will be performed as a desk-top study and will be based on collation of the required information to be used as input to the magnetic field study / analysis. The analysis will be supported by numerical (CDEGS) modelling and expression of findings as reflected against the applicable standards / guides and the requirements for magnetic field compliance. These are outlined in the Sections to follow.

Supporting information, used as input to the magnetic field study, is presented in Appendix A.

A brief motivation to the treatment of the 0,4 μ T level is covered in Appendix B.

An example of the energisation of a typical 3-phase-circuit is shown in Appendix C.

3.2 Applicable Standards / References

The magnetic field study will be guided by the following reference:

- [1] Rijksinstituut voor Volksgezondheid en Milieu (RIVM) (National Institute for Public Health and the Environment) - Ministry of Health, Welfare and Sport: Agreements on the Calculation Method for the "Magnetic Field Zone" for Below-Ground Cables and High-Voltage Substations, 3 November 2011.

Specific excerpts from these requirements are summarised in Table 1 below and shows how the study was shaped.

3.4 Guidelines of the Calculation Method

Table 1 summarises the some of the main requirements as reflected in [1].

Table 1: Summary of the main requirements for the Calculation Method as reflected in [1].

No	Requirement	Remarks
1	Calculations are performed for steady state and balanced conditions.	
2	Magnetic field levels to be calculated at 1 m above ground.	
3	The "open excavation" condition will be considered in this case with the cable cross section in the horizontal position with the cables in the cable bed.	See Appendix A for details.
4	Cable sleeves / armouring and excess lengths of the cables will not be taken into account.	Cables are modelled as conductors with insulated jackets.
5	The location of the 0,4 μT contour is of interest. The width of the magnetic field zone is then rounded according to the same rounding rules that are also laid down in the guide for overhead lines.	In this case the field zone will not be rounded but only the 0,4 μT contour shown in the deliverable, as requested by the client.
6	For take-off points, the study is bounded by the substation fence.	For modelling purposes, the cables linking with GTB and TBW may extend well outside the fenced area of OTD.
7	For the zone calculation, the ascending conductors from the station up to the first mast of a connected high-voltage line must be considered, even though these conductors can be (partially) outside the fence. Cables are included as far as they are inside the fence.	As noted in Section 3.5.4, the overhead lines are excluded from this study.
8	In cases where several stations are adjacent, they are considered as one station for the calculation. If stations are located in close proximity to each other but not directly adjacent, the magnetic field contour is calculated for each station separately. If there are two owners / network operators, both will have to be willing to exchange information about the magnetic field contour. If the different contours overlap, the envelope of both contours forms the magnetic field contour of the stations. Superposition of the magnetic fields is not taken into account. The same also applies to the points where the contour of the station overlaps with the magnetic field zone of the supply lines and cables that do not belong to the station. There, too, the envelope of both contours is maintained and superposition is not included.	As noted in Section 3.5.4, the overhead lines are excluded from this study.
9	For new / extended substations, the magnetic field contour should indicate the future end situation for a station when it is fully utilized.	See the energisations considered in Section 4 for the cases studied.

No	Requirement	Remarks
10	The calculation of the magnetic field contour is based on all current-carrying conductors. For a voltage levels of 150 kV, the calculation current is based on a single fault reserve (the n-1 criterion). This means that • For two circuits of the same voltage (150 kV), a calculation current of 50% of the design current is used. • For three circuits of the same connection and the same voltage (150 kV), a calculation current of 67 % of the design current is used. • For four circuits of the same connection and the same voltage (150 kV), a calculation current of 75 % of the design current is used.	In accordance with this statement, the following design values apply (from Appendix A): • Main busses = 3150 A • Cable fields = 2000 A • Coupling field = 3150 A • Transformer fields = 400 A The following energisation values will be used in the study: • Main busses (50%) = 1575 A • Cable fields (75 %): 1500 A • Coupling field (50 %): 1575 A • Transformer fields (67 %): 268 A It is noted that the current associated with the transformer fields (268 A per circuit) is conservative and above the maximum current levels expected based on the transformer ratings (See Appendix A).
11	The currents in the conductors of a circuit entering the station are assumed to be symmetrical.	
12	For live conductors of a circuit entering the station, the calculation assumes that the current direction in the conductors is always into the station.	
13	For conductors within the station, with the exception of the bus system, the current direction is assumed to be from high to low voltage.	
14	The conductors are assumed to carry current up to and including the furthest bus.	
15	With regard to the other components within the station, such as, - transformers, - current and voltage transformers - isolators, - oil-filled coils (which are encapsulated) - star point reactors are NOT included as these devices are not expected to contribute to the magnetic field outside the substation site boundary.	Only the transformer connections are modelled as start points to control current flow in the relevant circuits (see Section 3.5 for details).

3.5 Assumptions and Limitations to the Scope of Work

3.5.1 Interface Point

With reference to Figure 1, the interface points are as follows:

- The point between the primary and the secondary of the three Enexis transformers with only the 150 kV primary side (based on the new configuration) to be included in the study;
- The point of cable access to the new Oosteind 150 kV substation from Geertruidenberg (GTB150) and Tilburg West (TBW150) substations.

3.5.2 Cables Linking with Transformers

- The centre-point to centre-point of the cables in the trefoil formation was taken as 150 mm.

3.5.3 Cases Considered

The requirement for a dual busbar system, as in the case of the Oosteind substation, is to consider the following current flow configurations, in accordance with [1].

Bus	A	B
Current Flow Direction	↓	↓
	↓	↑
	↑	↓
	↑	↑

The current flow direction is specified by means of the current injection at the beginning of a circuit. Based on the requirements outlined in [1], the proposed cases considered in the study are covered in Section 4 of this document.

3.5.4 Exclusions from the Study

The following are excluded from the study based on the current scope of work.

- The part of the existing Oosteind substation that includes the 10 kV secondary side of the transformers, the 10 kV sections of the existing Oosteind substation and the overhead power lines associated with the existing Oosteind substation (see Figure 1).

It is noted that because of the proposed location of the new Oosteind substation, the magnetic fields presented by the overhead power lines may contribute to field levels of the new substation, particularly to the area south west of the existing substation. This may require further study based

on a revised scope of work but are not included in the present study based on the guidance of [1] (see Point 8 in Table 1).

- Mitigation of the magnetic field (by using passive loops, for example) where it exceeds 0,4 μT at certain locations is excluded from this study.

4. PROPOSED CASES TO BE STUDIED

4.1 General

Based on the Guidelines for the Calculation Method [1], the cases described in the Section 4.2 will be considered in the magnetic field study.

4.3 Cases Considered for the Magnetic Field Zone Calculation

The cases considered for the magnetic field zone calculation are based on all energised conductors [1] as presented in Appendix A, and included:

- All connections to GTB150 and TBW150 are energised (IN).

The system and configuration employed in the numerical model is summarised in Figure 2. The current flow on the busbar ss indicated in Table 2 and the assigned circuit currents shown in Table 3 below.

Table 2: Current flow on the busbar.

Bus	A	B
Current Flow Direction	↓	↓
	↓	↑
	↑	↓
	↑	↑

Table 3: Assigned circuit currents.

Circuit	Design Value	Value Used in Calculation	% of Design Value
Main busses	3150 A	1575 A	50 %
Cable fields	2000 A	1500 A	75 %
Coupling field	3150 A	1575 A	50 %
Transformer fields	400 A	268 A	67 %



Figure 2: System and configuration employed in the numerical model.

5. FINDINGS

The findings of the magnetic study will be addressed in this Section, once the study has been completed.

6. CONCLUDING REMARKS

6.1 Conclusions

The proposed plan of action (Plan van Aanpak) related to the new Oosteind 150 kV substation specific magnetic field zone study has been presented in this document.

6.2 Recommendations

It is recommended that the proposed Plan of Action covered in this document be submitted for review and approval by the client.

7. ACKNOWLEDGEMENT

Mr Wouter Mos, Lead Engineer – Large Projects, TenneT, is acknowledged and thanked for providing technical information related to the new Oosteind Substation.

Yours Sincerely

A handwritten signature in black ink, appearing to read 'Pieter H Pretorius', enclosed within a simple rectangular box.

Dr Pieter H Pretorius
Principal Consultant: Earthing & Electromagnetic Fields
Transmission & Distribution Technology
DNV GL - Energy Advisory

8. APPENDIX A – SUPPORTING INFORMATION TO THE MAGNETIC FIELD STUDY

8.1 Base Line Information Used as Input to the Study

The information captured in the Sections to follow was received from TenneT and was used as input to the development and execution of the magnetic field study. Section 8.2 through 8.10 covers general information about the substation layout and energization. Section 8.11 covers details pertaining to the cables.

8.2 Data - 150 kV Oosteind Substation

- Clock numbers: From A to B rail, 12, 4, 8, 12, 4, 8;
- Single line 150kV Oosteind - see Section 8.5;
- Plan of the primary layout with the station boundary - see Section 8.6;
- Layout of the high-voltage substation - see Section 8.7;
- Cross section of transformer fields - see Section 8.8;
- Cross section of coupling field - see Section 8.9;
- Cross-section of cable fields - see Section 8.5.10;

8.2 Data – Design Load per Field Type

- Main busses = 3150 A
- Cable fields = 2000 A
- Coupling field = 3150 A
- Transformer fields = 400 A

The field is suitable for this current level but considering the three transformers are rated at 63 MVA, 63 MVA and 44 MVA. Thus, 400 A will not be achieved in any situation. This is confirmed by using Equation A-1 to calculate the following currents associated with the power ratings: 63 MVA $\equiv$ 242,5 A and 44 MVA $\equiv$ 169,4 A.

$$S_{3\phi} = \sqrt{3}I_L V_L \quad \dots (A-1)$$

8.3 Data – Normal Operation

Once the new substation is installed, the 'normal' business operation will be as follows:

- The connections to Geertruidenberg (GTB150) will be energised (IN) and the connections to Tilburg West (TBW150) will be de-energised (OUT).
- During this normal situation, the current flow direction is from GTB150 to Oosteind (OTD150) and the power feed will be to the three Enexis transformers (63 MVA, 63 MVA and 44 MVA).

8.4 Data – Exceptional /Emergency Operation

During an emergency (disruption) situation, the following grid configuration will arise:

LEGENDA:

- Overspanningsafsluiter
- Aanzet
- Verenigingsafsluiter
- Schakelaar
- STroomafsluiter
- Spanningsratio
- Isolatieafsluiter
- Seriesinductor
- Waarheid
- Condensator

STATION OOSTIND 150

TRANSFORMATORVELD

KOPPELVELD

NEW STATION

RESERVE

GT Z

GT W

TBW Z

TBW W

RAIL A

RAIL B

RAIL C

RAIL D

RAIL E

RAIL F

RAIL G

RAIL H

RAIL I

RAIL J

RAIL K

RAIL L

RAIL M

RAIL N

RAIL O

RAIL P

RAIL Q

RAIL R

RAIL S

RAIL T

RAIL U

RAIL V

RAIL W

RAIL X

RAIL Y

RAIL Z

RAIL AA

RAIL AB

RAIL AC

RAIL AD

RAIL AE

RAIL AF

RAIL AG

RAIL AH

RAIL AI

RAIL AJ

RAIL AK

RAIL AL

RAIL AM

RAIL AN

RAIL AO

RAIL AP

RAIL AQ

RAIL AR

RAIL AS

RAIL AT

RAIL AU

RAIL AV

RAIL AW

RAIL AX

RAIL AY

RAIL AZ

RAIL BA

RAIL BB

RAIL BC

RAIL BD

RAIL BE

RAIL BF

RAIL BG

RAIL BH

RAIL BI

RAIL BJ

RAIL BK

RAIL BL

RAIL BM

RAIL BN

RAIL BO

RAIL BP

RAIL BQ

RAIL BR

RAIL BS

RAIL BT

RAIL BU

RAIL BV

RAIL BW

RAIL BX

RAIL BY

RAIL BZ

RAIL CA

RAIL CB

RAIL CC

RAIL CD

RAIL CE

RAIL CF

RAIL CG

RAIL CH

RAIL CI

RAIL CJ

RAIL CK

RAIL CL

RAIL CM

RAIL CN

RAIL CO

RAIL CP

RAIL CQ

RAIL CR

RAIL CS

RAIL CT

RAIL CU

RAIL CV

RAIL CW

RAIL CX

RAIL CY

RAIL CZ

RAIL DA

RAIL DB

RAIL DC

RAIL DD

RAIL DE

RAIL DF

RAIL DG

RAIL DH

RAIL DI

RAIL DJ

RAIL DK

RAIL DL

RAIL DM

RAIL DN

RAIL DO

RAIL DP

RAIL DQ

RAIL DR

RAIL DS

RAIL DT

RAIL DU

RAIL DV

RAIL DW

RAIL DX

RAIL DY

RAIL DZ

RAIL EA

RAIL EB

RAIL EC

RAIL ED

RAIL EE

RAIL EF

RAIL EG

RAIL EH

RAIL EI

RAIL EJ

RAIL EK

RAIL EL

RAIL EM

RAIL EN

RAIL EO

RAIL EP

RAIL EQ

RAIL ER

RAIL ES

RAIL ET

RAIL EU

RAIL EV

RAIL EW

RAIL EX

RAIL EY

RAIL EZ

RAIL FA

RAIL FB

RAIL FC

RAIL FD

RAIL FE

RAIL FF

RAIL FG

RAIL FH

RAIL FI

RAIL FJ

RAIL FK

RAIL FL

RAIL FM

RAIL FN

RAIL FO

RAIL FP

RAIL FQ

RAIL FR

RAIL FS

RAIL FT

RAIL FU

RAIL FV

RAIL FW

RAIL FX

RAIL FY

RAIL FZ

RAIL GA

RAIL GB

RAIL GC

RAIL GD

RAIL GE

RAIL GF

RAIL GG

RAIL GH

RAIL GI

RAIL GJ

RAIL GK

RAIL GL

RAIL GM

RAIL GN

RAIL GO

RAIL GP

RAIL GQ

RAIL GR

RAIL GS

RAIL GT

RAIL GU

RAIL GV

RAIL GW

RAIL GX

RAIL GY

RAIL GZ

RAIL HA

RAIL HB

RAIL HC

RAIL HD

RAIL HE

RAIL HF

RAIL HG

RAIL HH

RAIL HI

RAIL HJ

RAIL HK

RAIL HL

RAIL HM

RAIL HN

RAIL HO

RAIL HP

RAIL HQ

RAIL HR

RAIL HS

RAIL HT

RAIL HU

RAIL HV

RAIL HW

RAIL HX

RAIL HY

RAIL HZ

RAIL IA

RAIL IB

RAIL IC

RAIL ID

RAIL IE

RAIL IF

RAIL IG

RAIL IH

RAIL II

RAIL IJ

RAIL IK

RAIL IL

RAIL IM

RAIL IN

RAIL IO

RAIL IP

RAIL IQ

RAIL IR

RAIL IS

RAIL IT

RAIL IU

RAIL IV

RAIL IW

RAIL IX

RAIL IY

RAIL IZ

RAIL JA

RAIL JB

RAIL JC

RAIL JD

RAIL JE

RAIL JF

RAIL JG

RAIL JH

RAIL JI

RAIL JJ

RAIL JK

RAIL JL

RAIL JM

RAIL JN

RAIL JO

RAIL JP

RAIL JQ

RAIL JR

RAIL JS

RAIL JT

RAIL JU

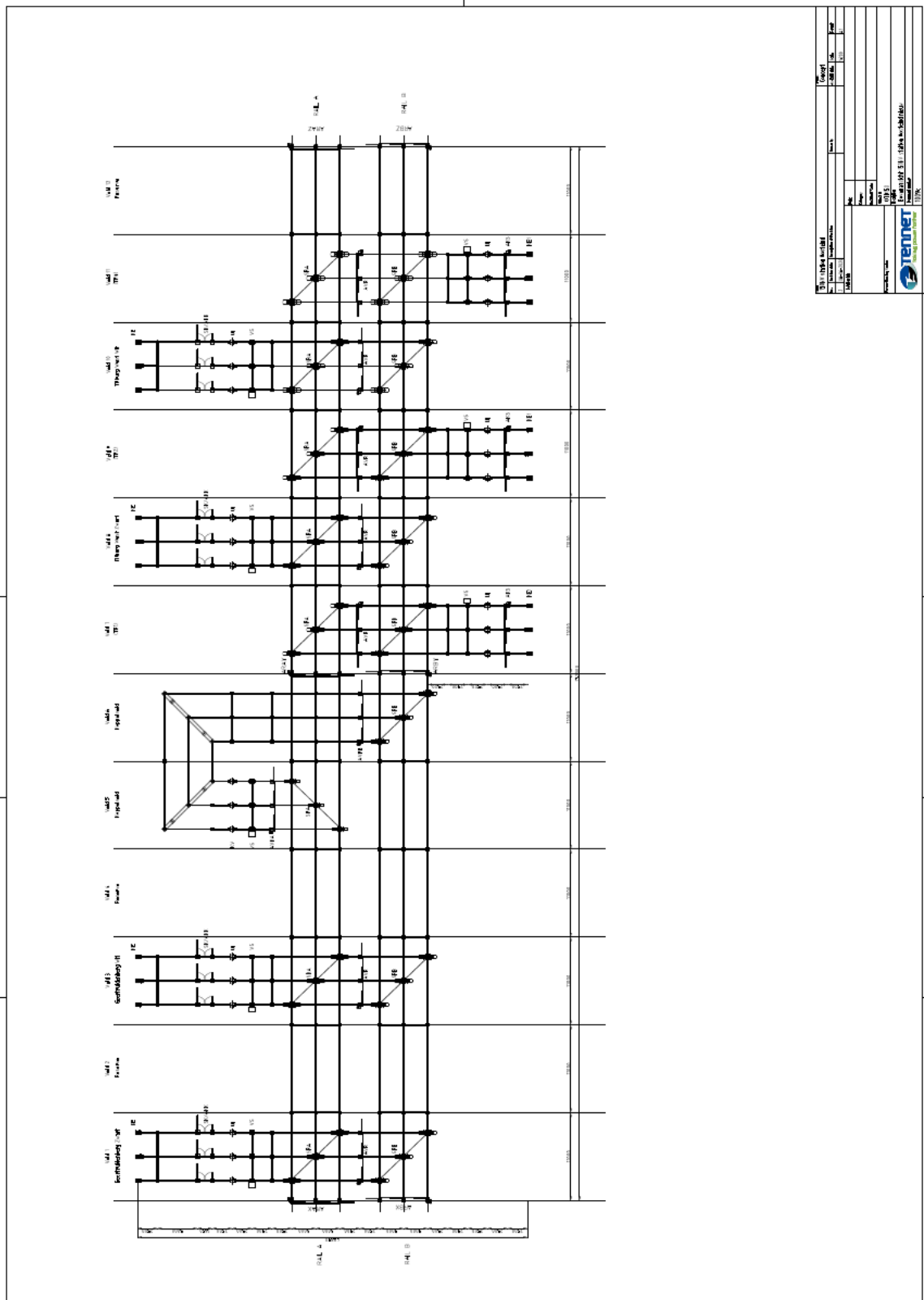
RAIL JV

RAIL JW

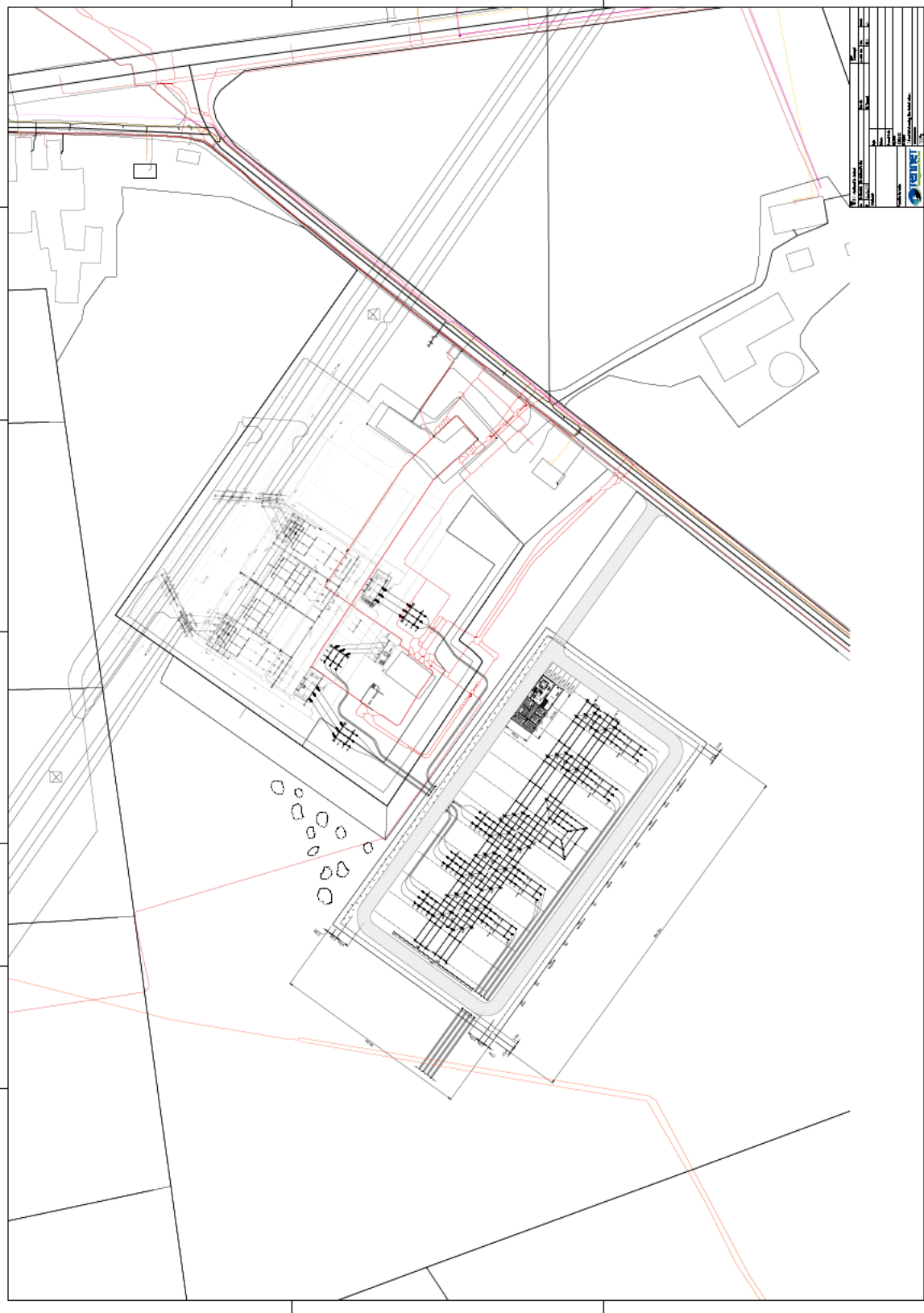
RAIL JX

RAIL

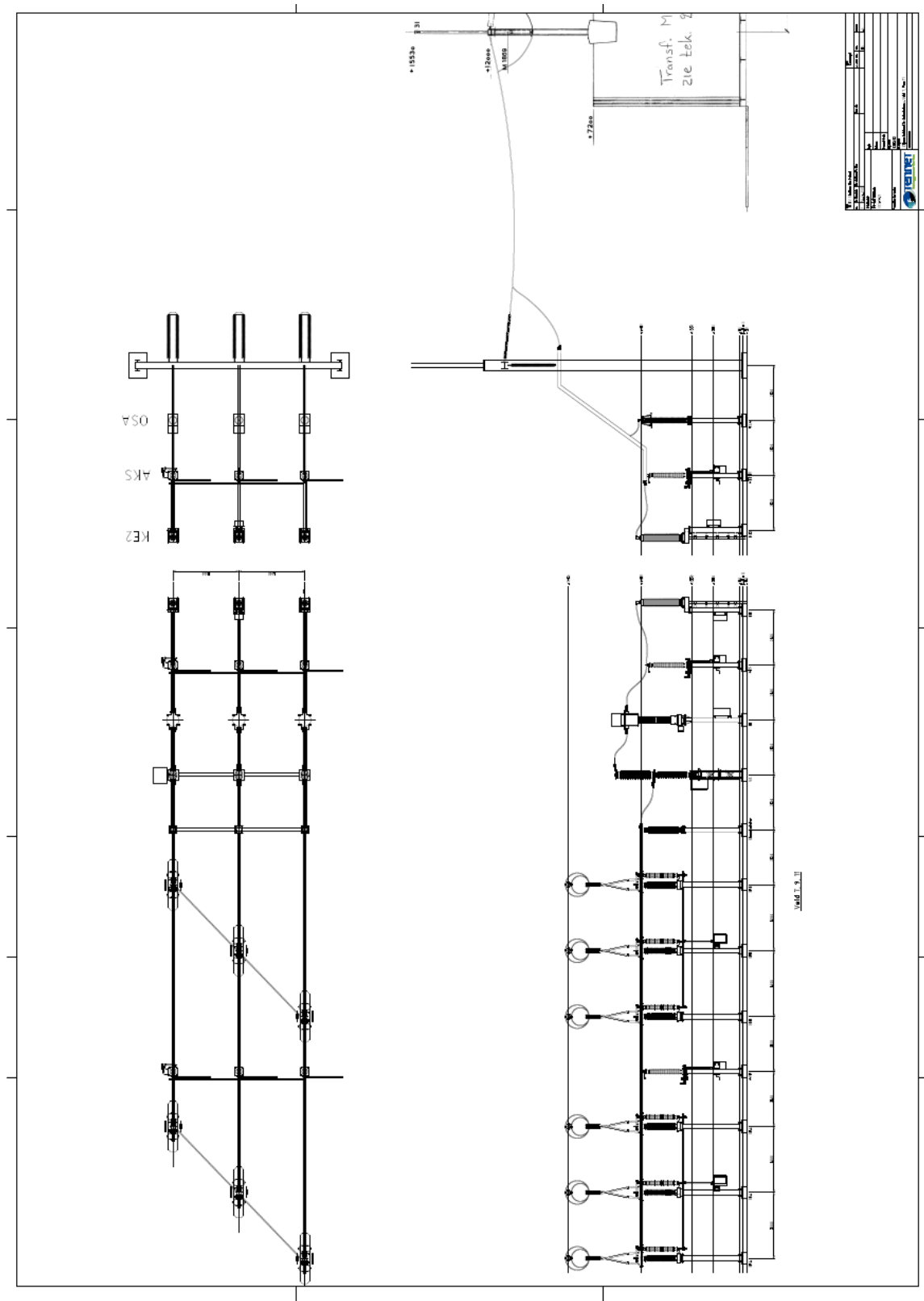
8.6 Plan of the Primary Layout with the Station Boundary

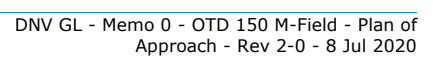


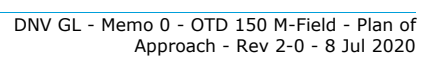
8.7 Layout of the High Voltage Substation



8.8 Cross Section of the Transformer Fields



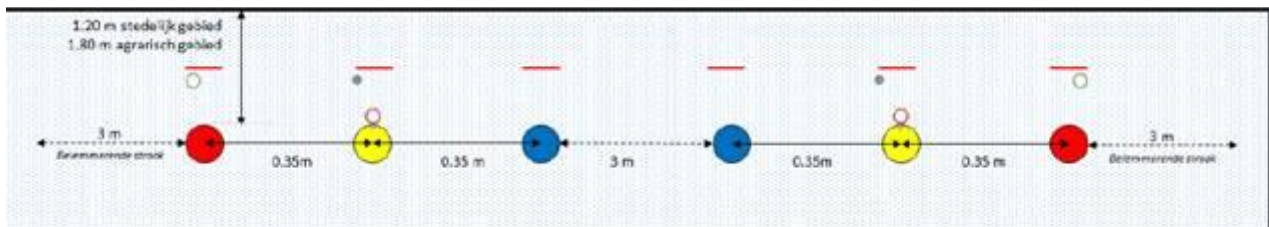




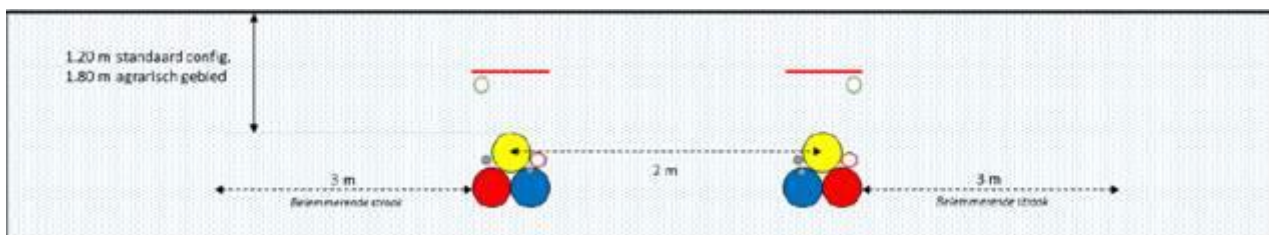
8.11 Data - Cables

In view of the cable sections at the new Oosteind 150kV substation, the following applies:

- Cables between 150 kV Oosteind – 150 kV Geertruidenberg and 150 kV Oosteind - 150kV Tilburg West:
 - These cables are buried at a depth of 1,8m.
 - The phases of each circuit are 0,35 m apart and the circuits themselves are 3 m apart as per the image below.
 - To avoid confusion, it is possible that the 150 kV cables may run underneath the bypass for modelling purposes.



- Cables between 150 kV Oosteind and the three existing Enexis transformers:
 - These cables are buried at a depth of 1,8 m.
 - The phases of each circuit are adjacent to each other and the circuits themselves are at least 2 m apart as per the image below.



9. APPENDIX B – SUMMARY ON THE MOTIVATION OF EMPLOYING THE 0,4 μ T LEVEL

9.1 Magnetic Fields and Health

Similar to electric appliances, high voltage power systems generate magnetic fields. Of interest is the 50 Hz alternating fields in the vicinity of these systems.

It is possible for very high, 50 Hz, magnetic fields to cause excitation of the nervous system that may lead to muscle contraction. Such very high magnetic field levels are not presented by typical power systems and installations found in the conventional living and working environment.

At lower field levels, but above a certain threshold value, it is possible that magnetic fields may cause a flickering visual sensation, known as phosphenes, and considered an acute effect. This effect is not harmful and is also very rarely associated with typical power systems and their working or living environments. Using this threshold value as a guide, the European Commission has recommended a reference level of 100 microtesla, as 50 Hz magnetic field exposure level for members of the public. Below this reference level, the magnetic field does not cause acute effects.

Some debate exists on the possible effects of long-term exposure to low level magnetic fields (field levels well below the reference level). Some research near overhead power lines suggested that children living close to power lines may have an increased risk of developing leukemia. This suggestion is linked with long-term exposure to magnetic fields, on average, around 0,4 microtesla. A causal relationship between magnetic fields and childhood leukemia, however, has not been established. Recent studies from Denmark and the United Kingdom do not support an increased health risk. Drawing a firm concluding from scientific research (in general and in retrospect), that children living near high-voltage lines or children that stays near high voltage lines for a long period of time, may experience an increased health risk, is viewed as limited.

9.2 Why Calculate the 0,4 microtesla Zone?

The Ministry of Infrastructure and the Environment follows a precautionary-policy based on the recommended exposure value of 0,4 microtesla for new situations involving “sensitive” locations, such as, homes, schools and places of childcare that fall close to overhead power lines.

This policy includes an established calculation method for the calculation of the specific magnetic field zone. The specific magnetic field zone is the area around the high-voltage line within which the calculated annual average magnetic field strength is higher than 0,4 microtesla.

9.3 Zone Calculation

The guidance from [1] defines the 'specific magnetic field zone': this is the zone where the magnetic field level is, on average, higher than 0,4 microtesla per year, or that may become so in the future. The way in which this specific magnetic field zone can be calculated is presented in a guide managed by RIVM [1]. DNV GL has been designated as “an entity known to have experience with zone calculations according to the guide”.

In order to translate the uncertainty about the possible health effects into a concrete zone calculation, certain choices and simplifications have been made in the noted guide. Simplifications are inevitable because the full characteristics of the current are not always and everywhere known in the high voltage network. These simplifications lead to the fact that the specific magnetic field zone

calculated in this report does not reflect the actual strength of the magnetic field at a specific location at a specific time, but rather a magnetic field zone that fits the government's high voltage substation and cable policy [1].

9.4 Disclaimer

The 0,4 μT magnetic field contours reflected in this report were calculated for a part of the high voltage grid. The calculations were done in accordance with the RIVM guide [1]. The fact that 0,4 microtesla zones and contours have been calculated in this report does not mean that an increased health risk is expected within these zones. The 0,4 microtesla zones indicate within which distance from the high-voltage connection it is recommended to avoid the development of new sensitive locations, provided that the high voltage connection consists of the infrastructure covered in this report.

10. APPENDIX C – EXAMPLE ENERGISATION

An example of the energisation of a typical 3-phase-circuit, using the Lead Energization in CDEGS, is shown in this Appendix C.

Figure 10-1 shows the 3 phases with Lead Excitation (injection) from the left and Lead Excitation (extraction) from the right of the three-phase circuit (2 conductors). The energization is specified as shown in Table 10-1.

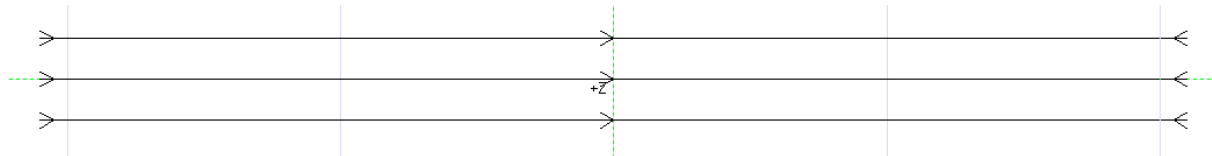


Figure 10-1.

Table 10-1.

Bus	Identification	Energization Type	Magnitude	Angle
1	PH-A-IN	Current (Lead)	1000	0
2	PH-A-OUT	Current (Lead)	-1000	0
3	PH-B-IN	Current (Lead)	1000	120
4	PH-B-OUT	Current (Lead)	-1000	120
5	PH-C-IN	Current (Lead)	1000	-120
6	PH-C-OUT	Current (Lead)	-1000	-120

OR

Bus	Identification	Energization Type	Magnitude	Angle
1	PH-A-IN	Current (Lead)	1000	0
2	PH-A-OUT	Current (Lead)	1000	180
3	PH-B-IN	Current (Lead)	1000	120
4	PH-B-OUT	Current (Lead)	1000	-60
5	PH-C-IN	Current (Lead)	1000	-120
6	PH-C-OUT	Current (Lead)	1000	60

Figure 10-2 shows the noted energization schematically with current flow direction indicated.

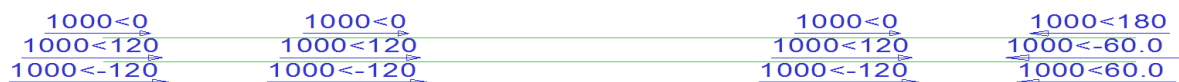


Figure 10-2.